

Nodal Analysis

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SENSE

Steps to Analyse AC circuits

1. Transform the circuit to the phasor or frequency domain
2. Solve the problem using circuit techniques (nodal analysis, mesh analysis, or any theorems)
3. Transform the resulting phasor to the time domain.

Nodal Analysis

- Nodal analysis uses node voltages as the circuit variables.

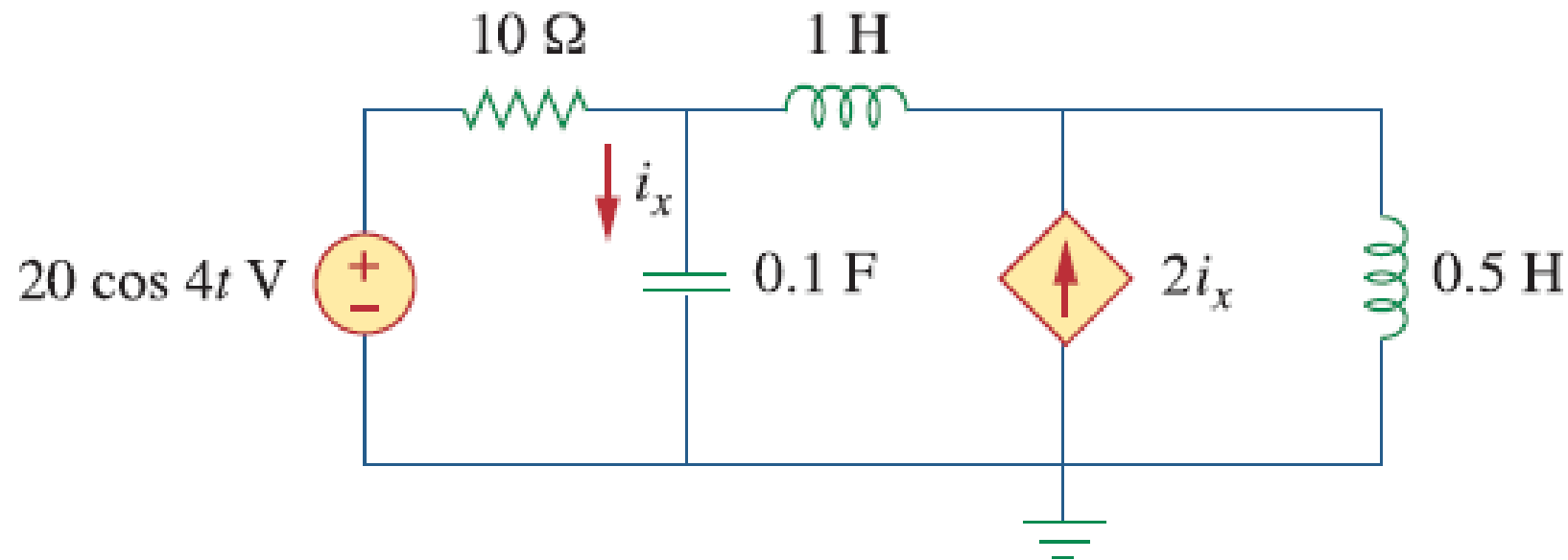
Steps to Determine Node Voltages:

1. Select a node as the reference node. Assign voltages $v_1, v_2, \dots, v_{n-1}$ to the remaining $n - 1$ nodes. The voltages are referenced with respect to the reference node.
2. Apply KCL to each of the $n - 1$ nonreference nodes. Use Ohm's law to express the branch currents in terms of node voltages.
3. Solve the resulting simultaneous equations to obtain the unknown node voltages.

Nodal Analysis

Numerical - 1

Find i_x in the circuit using nodal analysis.



Solution:

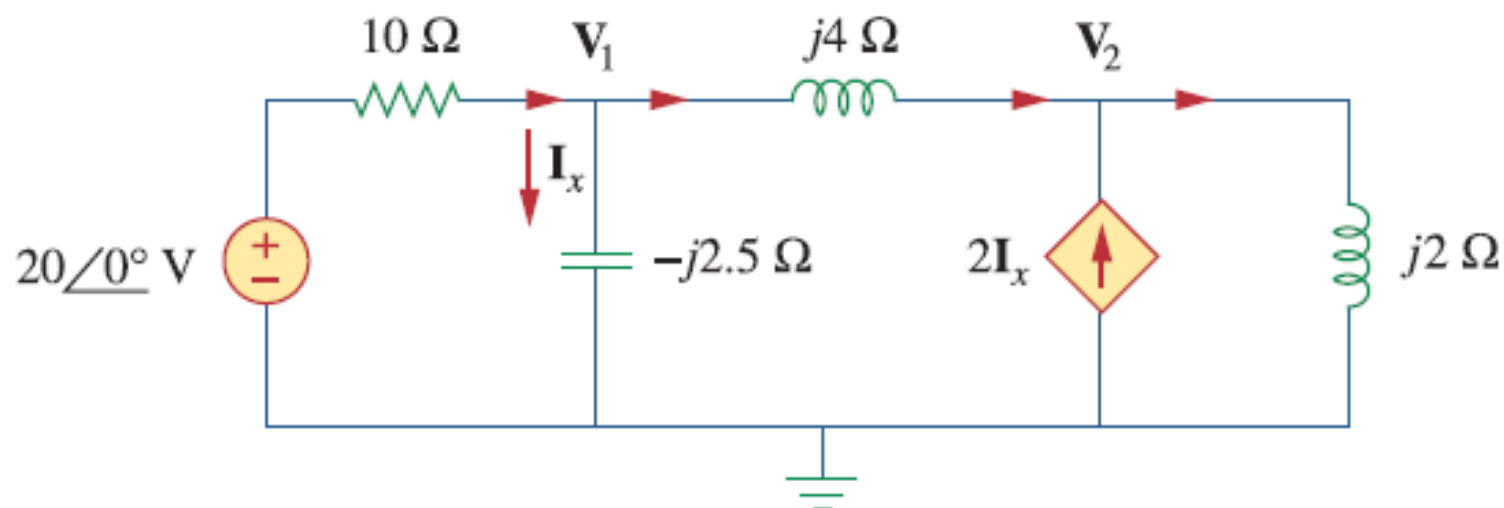
We first convert the circuit to the frequency domain:

$$20 \cos 4t \Rightarrow 20 \angle 0^\circ, \quad \omega = 4 \text{ rad/s}$$

$$1 \text{ H} \Rightarrow j\omega L = j4$$

$$0.5 \text{ H} \Rightarrow j\omega L = j2$$

$$0.1 \text{ F} \Rightarrow \frac{1}{j\omega C} = -j2.5$$



Frequency domain equivalent of the circuit

Applying KCL at node 1,

$$\frac{20 - \mathbf{V}_1}{10} = \frac{\mathbf{V}_1}{-j2.5} + \frac{\mathbf{V}_1 - \mathbf{V}_2}{j4}$$

$$(1 + j1.5)\mathbf{V}_1 + j2.5\mathbf{V}_2 = 20 \longrightarrow (1)$$

At node 2,

$$2\mathbf{I}_x + \frac{\mathbf{V}_1 - \mathbf{V}_2}{j4} = \frac{\mathbf{V}_2}{j2}$$

But $\mathbf{I}_x = \mathbf{V}_1 / -j2.5$. Substituting this gives

$$\frac{2\mathbf{V}_1}{-j2.5} + \frac{\mathbf{V}_1 - \mathbf{V}_2}{j4} = \frac{\mathbf{V}_2}{j2}$$

By simplifying, we get

$$11\mathbf{V}_1 + 15\mathbf{V}_2 = 0 \longrightarrow (2)$$

Equation (1) and (2) can be put in matrix form as

$$\begin{bmatrix} 1 + j1.5 & j2.5 \\ 11 & 15 \end{bmatrix} \begin{bmatrix} \mathbf{V}_1 \\ \mathbf{V}_2 \end{bmatrix} = \begin{bmatrix} 20 \\ 0 \end{bmatrix}$$

$$\Delta = \begin{vmatrix} 1 + j1.5 & j2.5 \\ 11 & 15 \end{vmatrix} = 15 - j5$$

$$\Delta_1 = \begin{vmatrix} 20 & j2.5 \\ 0 & 15 \end{vmatrix} = 300$$

$$\Delta_2 = \begin{vmatrix} 1 + j1.5 & 20 \\ 11 & 0 \end{vmatrix} = -220$$

$$\mathbf{V}_1 = \frac{\Delta_1}{\Delta} = \frac{300}{15 - j5} = 18.97 \angle 18.43^\circ \text{ V}$$

$$\mathbf{V}_2 = \frac{\Delta_2}{\Delta} = \frac{-220}{15 - j5} = 13.91 \angle 198.3^\circ \text{ V}$$

The current $\mathbf{I}_x$ is given by

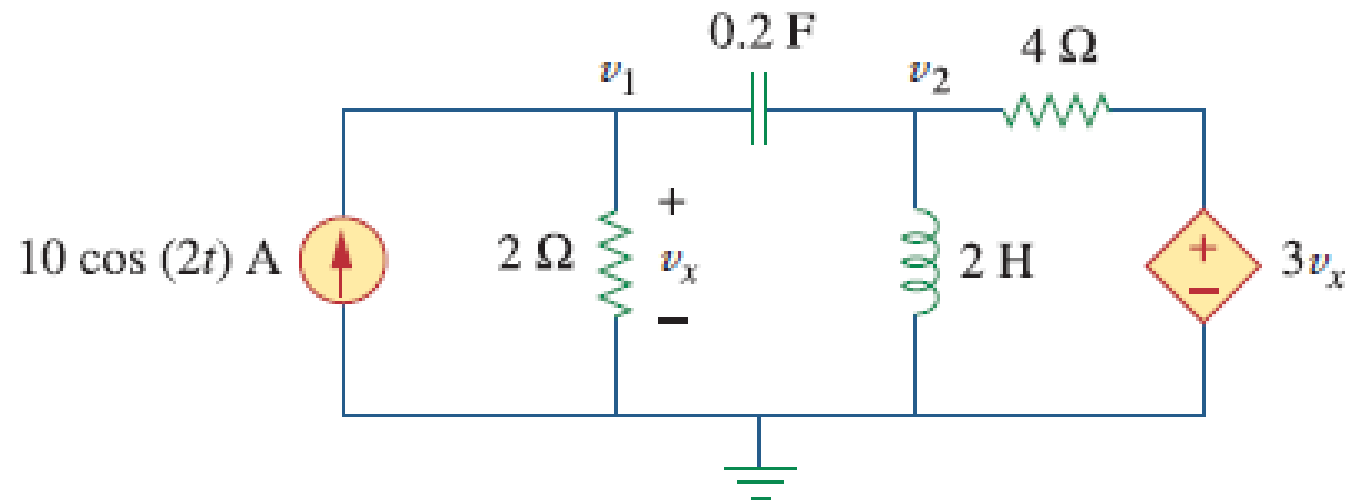
$$\mathbf{I}_x = \frac{\mathbf{V}_1}{-j2.5} = \frac{18.97 \angle 18.43^\circ}{2.5 \angle -90^\circ} = 7.59 \angle 108.4^\circ \text{ A}$$

Transforming this to the time domain,

$$i_x = 7.59 \cos(4t + 108.4^\circ) \text{ A}$$

- Home work Problem -1:

Using nodal analysis, find v_1 and v_2 in the circuit



Answer: $v_1(t) = 11.325 \cos(2t + 60.01^\circ) \text{ V}$,
 $v_2(t) = 33.02 \cos(2t + 57.12^\circ) \text{ V}$.